

# MATH 180 Graph Theory — Discussion 10

## Indicator variables and linearity of expectation (Ch. 10.3)

### Quick Recall (Ch. 10.3)

**Random variable & expectation.** Let  $(\Omega, P)$  be a finite probability space. A *random variable* is a function  $f : \Omega \rightarrow \mathbb{R}$ . Its *expectation* is

$$\mathbb{E}[f] = \sum_{\omega \in \Omega} P(\{\omega\}) \cdot f(\omega).$$

When  $P$  is uniform on  $\Omega$ , this is just the average value  $\mathbb{E}[f] = \frac{1}{|\Omega|} \sum_{\omega} f(\omega)$ .

**Indicator of an event.** For an event  $A \subseteq \Omega$ , the *indicator*  $I_A$  is the random variable

$$I_A(\omega) = \begin{cases} 1 & \omega \in A, \\ 0 & \omega \notin A. \end{cases}$$

*Fact.*  $\mathbb{E}[I_A] = P(A)$ .

**Linearity of expectation.** For random variables  $f, g$  on the same  $(\Omega, P)$  and any  $\alpha \in \mathbb{R}$ ,

$$\mathbb{E}[\alpha f + g] = \alpha \mathbb{E}[f] + \mathbb{E}[g].$$

*Independence is not required.*

**Random coin-toss space  $C_n$ .**  $\Omega = \{0, 1\}^n$  with the uniform distribution, modeling  $n$  independent fair coin tosses (write 0 = T, 1 = H).

**Mini-example (lecture).** Let  $f_1 : C_n \rightarrow \mathbb{R}$  count the number of 1's. Setting  $A_i = \{s \in C_n : s_i = 1\}$ , we have  $f_1 = \sum_{i=1}^n I_{A_i}$  and  $P(A_i) = 1/2$ , so by linearity

$$\mathbb{E}[f_1] = \sum_{i=1}^n P(A_i) = \frac{n}{2}.$$

### Practice: Textbook Exercise 10.3.6\*

**Exercise 10.3.6\*.** We toss a fair coin  $n$  times. What is the expected number of “runs”? Runs are consecutive tosses with the same result. For instance, the toss sequence HHHTTHTH has 5 runs.

#### Hint

**Hint (textbook).** It is better to count *boundaries* among runs. The probability that a given position between two tosses is a boundary is  $1/2$ .

**Answer: Step 1: Re-frame using boundaries.** A *boundary* is a position between two consecutive tosses that yield different results. In a sequence of  $n$  tosses, the number of runs equals the number of boundaries plus one:

$$\# \text{ runs} = 1 + \# \text{ boundaries}.$$

(Each boundary separates one run from the next, so a sequence with  $k$  boundaries decomposes into exactly  $k + 1$  runs.) For example, in HHHTTHTH ( $n = 8$ ): boundaries between positions 3–4 (H→T), 5–6 (T→H), 6–7 (H→T), 7–8 (T→H), giving 4 boundaries and  $1 + 4 = 5$  runs. ✓

**Step 2: Set up indicators.** For each  $i \in \{1, 2, \dots, n-1\}$ , let

$$B_i = \{s \in C_n : s_i \neq s_{i+1}\}$$

be the event “position between toss  $i$  and toss  $i + 1$  is a boundary.” Then

$$\# \text{ boundaries} = \sum_{i=1}^{n-1} I_{B_i}, \quad \text{so} \quad \# \text{ runs} = 1 + \sum_{i=1}^{n-1} I_{B_i}.$$

**Step 3: Compute  $P(B_i)$ .** Tosses  $i$  and  $i + 1$  are independent and uniform on  $\{0, 1\}$ , so the four outcomes  $(s_i, s_{i+1}) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$  are equally likely. Two of them have  $s_i \neq s_{i+1}$ , hence

$$P(B_i) = \frac{2}{4} = \frac{1}{2} \quad \text{for every } i \in \{1, \dots, n-1\}.$$

**Step 4: Apply linearity of expectation.** Since  $\mathbb{E}[I_{B_i}] = P(B_i) = 1/2$ , linearity gives

$$\mathbb{E}[\# \text{ runs}] = \mathbb{E}\left[1 + \sum_{i=1}^{n-1} I_{B_i}\right] = 1 + \sum_{i=1}^{n-1} \mathbb{E}[I_{B_i}] = 1 + (n-1) \cdot \frac{1}{2}.$$

That is,

$$\mathbb{E}[\# \text{ runs}] = \frac{n+1}{2}.$$

□

*Sanity check (small  $n$ ).*

- $n = 1$ : only one toss, always 1 run. Formula:  $(1+1)/2 = 1$ . ✓
- $n = 2$ : outcomes HH, HT, TH, TT with 1, 2, 2, 1 runs respectively. Average =  $(1+2+2+1)/4 = 3/2 = (2+1)/2$ . ✓
- $n = 8$ : formula gives  $9/2 = 4.5$ . The example sequence HHHTTHTH has 5 runs — slightly above average, as expected.

*Remark (contrast with the left-maxima example).* The lecture computed  $\mathbb{E}[f_2] = H_n = \sum_{i=1}^n 1/i$  for  $f_2$  = number of left maxima of a random permutation  $\pi \in S_n$ . Both proofs use the same toolkit: write the quantity of interest as a sum of indicators, then apply linearity. The crucial difference is in the indicator probabilities:

- For left maxima,  $P(\text{position } i \text{ is a left max}) = 1/i$  — *position-dependent*, yielding the harmonic sum  $H_n \sim \ln n$ .
- For boundaries,  $P(\text{position } i \text{ is a boundary}) = 1/2$  — *uniform across positions*, yielding the linear sum  $(n-1)/2$ .

Choosing the right indicators (here: boundaries rather than runs themselves) is the technique's main artistic step.